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Dynamic Scheduling Optimization Method for Flexible Job Shop Based on Digital Twin

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Abstract: With the in-depth implementation of the "Industry 4.0" and "Dual Carbon" strategies, the manufacturing of reducer boxes is accelerating its transformation towards intelligence and greenization. To address the frequent dynamic disturbances such as machine failures and urgent order insertions in actual production, as well as the difficulty for traditional scheduling methods to balance production efficiency and green energy saving, a green dynamic scheduling optimization method for flexible job shop driven by digital twin is proposed. First, a digital twin dynamic scheduling framework comprising a physical workshop, a virtual workshop, and a service system is constructed, and a high-fidelity simulation model of the reducer box flexible production line is built based on AnyLogic. Second, a multi-objective dynamic scheduling mathematical model is established by comprehensively considering the makespan, energy consumption, and rescheduling machine deviation. An Improved Multi-Objective Artificial Bee Colony (IMOABC) algorithm is designed to solve the problem, which enhances the global exploration and local exploitation capabilities by fusing Improved Precedence Operation Crossover (IPOX), uniform crossover strategies, and a variable step-size neighborhood search mechanism. Finally, multi-dimensional comparative validation is conducted based on standard benchmark instances and a reducer box manufacturing case. The results demonstrate that the proposed method can effectively cope with dynamic disturbances and outperforms traditional scheduling strategies in shortening production cycles, reducing equipment energy consumption, and maintaining system stability.

Keywords: Flexible job shop scheduling; Multi-objective optimization; Digital twin simulation; Multi-objective artificial bee colony algorithm; AnyLogic

1 Introduction

With the in-depth advancement of the "Industry 4.0" and "Dual Carbon" strategies, the manufacturing of reducer boxes is accelerating toward intelligence and greenization, and its production process exhibits typical characteristics of the Multi-Objective Flexible Job Shop Scheduling Problem (MOFJSP). Early studies mainly focused on time efficiency optimization under ideal static environments; however, the actual production shop is full of dynamic uncertainty, and disturbance events such as machine failures and urgent order insertions occur frequently, often leading to the failure of preset scheduling schemes. For machine failure disturbances, He and Sun [1] introduced multiple strategies combining idle time insertion, route change, and right-shift, and enhanced the robustness and stability of the system in response to machine failures by using a rescheduling algorithm. For the impact of urgent order insertions, He et al. [2] established a multi-constraint model and achieved rapid response to dynamic orders through the Non-dominated Sorting Genetic Algorithm III. In addition, to meet the requirements of green manufacturing, Küster et al. [3] analyzed the power differences of equipment states and proposed a multi-objective collaborative optimization method to simultaneously consider makespan and energy consumption. Liu et al. [4] developed a deep reinforcement learning-based hierarchical distributed framework for dynamic flexible job shop scheduling to enable real-time decision-making under continuous job arrivals. Luo [5] investigated the dynamic flexible job shop scheduling problem with new job insertions and proposed a deep Q-network method to achieve adaptive dispatching in real time.

Facing the high-dimensional Nondeterministic Polynomial-time hard problem with the coupling of energy consumption and dynamic disturbances, traditional algorithms are prone to falling into local optima. Therefore, researchers first focused on algorithm mechanism innovation. Mahmud et al. [6] designed a hybrid evolutionary

algorithm based on switching strategies, balancing global exploration and local exploitation capabilities. Wang et al. [7] proposed an Artificial Bee Colony algorithm, combined with multi-strategy initialization and a local search based on the critical path, significantly optimizing the makespan of flexible job shop scheduling. Liu et al. [8] combined Digital Twin to construct an adaptive scheduling mechanism, further improving the ability of the algorithm to cope with processing fluctuations. At the same time, Digital Twin technology provides a new paradigm for the closed-loop verification of scheduling schemes. Sun et al. [9] designed an intelligent manufacturing system framework based on Digital Twin, realizing real-time monitoring of shop floor states. Liu et al. [10], for the scheduling optimization problem under the multi-variety small-batch production mode, constructed a "feature–process–machine tool" hypernetwork model based on the Digital Twin workshop, realizing centralized management of multi-type data and rapid and efficient generation of intelligent scheduling schemes. Wang and Wu [11], aiming at the problem that uncertainty factors affect shop scheduling, constructed a planning and scheduling system based on Digital Twin, realizing the control of uncertainty factors in production activities and the guidance of actual production. Asorose and Adams [12] integrated Lean Six Sigma methods with Digital Twin technology, constructed a data-driven decision system, and realized predictive optimization and efficiency improvement of supply chain and production operation processes. Chen et al. [13] proposed a digital twin-oriented collaborative optimization method for fuzzy flexible job shop scheduling under multiple uncertainties, considering processing time, due date, and maintenance cycle simultaneously. Luo et al. [14] proposed a deep reinforcement learning-based dynamic multi-objective scheduling framework for flexible job shops to handle disturbances such as new job insertions and machine breakdowns. Caldeira et al. [15] studied a MOFJSP considering new job arrivals and energy consumption, and solved it using an improved backtracking search algorithm.

In view of this, this paper proposes a joint optimization method for green dynamic flexible job shop driven by Digital Twin. First, a multi-objective dynamic scheduling model is established by comprehensively considering makespan, energy consumption, and machine deviation. Second, an Improved Multi-Objective Artificial Bee Colony (IMOABC) algorithm integrating Improved Precedence Operation Crossover (IPOX) and variable step-size search is designed to solve the problem. Finally, a workshop Digital Twin system is constructed based on AnyLogic to realize the dynamic verification of scheduling schemes.

2 Problem Description and Model Establishment

2.1 Description of Dynamic Scheduling Problem in Workshop

Aiming at the deficiency that traditional MOFJSP studies often ignore dynamic disturbances, this paper focuses on the scheduling optimization problem in workshops under random events such as machine failures and urgent order insertions. This problem can be described as: under the premise of satisfying given process constraints, n reducer box workpieces set $\{P_1, P_2, \ldots, P_n\}$ and their corresponding sequence of J_i operations $\{O_{i1}, O_{i2}, \ldots, O_{iJ_i}\}$ are reasonably assigned to m processing machines set $\{M_1, M_2, \ldots, M_m\}$. Due to varying equipment performance, identical operations differ greatly in processing time and energy consumption across machines. Against this backdrop, a rescheduling strategy is adopted to timely revise and reoptimize the initial scheduling scheme under dynamic disturbances, reducing disturbance impacts and ensuring efficient workshop production.

2.2 Establishment of Dynamic Scheduling Model in Workshop

The relevant symbols of the model and their meanings are shown in Table 1.

Considering the complexity of the actual production system, the model is simplified as follows without loss of generality.

(1) All workpieces and machines are available at the initial time, and each machine can only process one task at the same time.

(2) Except for passive interruption caused by machine failure, the operation process is continuous and indivisible.

(3) The energy consumption model only considers processing energy consumption and idle energy consumption under steady state, ignoring transient fluctuations.

(4) For dynamic events, the executed scheduling scheme before time T_{d1} or T_{d2} is set as rigid constraints and remains unchanged, and only subsequent operations are reconstructed.

(5) It is assumed that the algorithm response is real-time, and the computational time cost of rescheduling decision is ignored.

This paper considers makespan, energy consumption, and machine deviation to construct the objective functions and establish constraints:

(1) Makespan $C_{\max}$ is a key indicator to evaluate workshop production efficiency. The objective function of minimizing makespan is f_1, that is:

$$f_1 = \min C_{\max} = \min\left(\max\left\{JF_i \mid \mathrm{i} = 1, 2, \cdots, \mathrm{n}\right\}\right) \tag{1}$$

Table 1. Symbols and definitions of the mathematical model

Indices	**Description**
i	Index of workpiece
j	Index of operation
k	Index of machine
Parameters	**Description**
n	Number of workpieces
m	Number of machines
J_i	Number of operations of workpiece i
O_{ij}	The j-th operation of workpiece i
Variables	**Description**
JF_i	Completion time of workpiece i
MF_k	Completion time of machine k
M_{ij}	Set of alternative machines for operation O_{ij}
MS_l^k	Start processing time of the l-th operation on machine k
MF_l^k	Finish processing time of the l-th operation on machine k
S_{ijk}	Start processing time of operation O_{ij} on machine k
F_{ijk}	Finish processing time of operation O_{ij} on machine k
P_{ijk}	Processing time of operation O_{ij} on machine k
WP_k	Unit processing energy consumption of machine k
IP_k	Unit idle energy consumption of machine k
$P(X_{ijk})$	Probability that operation O_{ij} is processed on machine k
T_{d1}	Time of machine failure occurrence
M_k	Failed machine
T_r	Expected repair time
$O_{i^*j^*}$	Operation being processed when machine failure occurs
Ω_{insert}	Set of urgent inserted orders
Md_{ij}	Machine change status of the j-th operation of workpiece i
MS_{ij}	Processing machine of the j-th operation of workpiece i in the original scheduling scheme
MS'_{ij}	Processing machine of the j-th operation of workpiece i in the rescheduling scheme
Decision Variables	**Description**
P_k	Decision variable, machine k is in processing state, $P_k = 1$, otherwise $P_k = 0$
X_{ijk}	Decision variable, operation O_{ij} is processed on machine k, $X_{ijk} = 1$, otherwise $X_{ijk} = 0$

(2) Total energy consumption E_{all} can be divided into processing energy consumption E_{busy} when the machine is in processing state and idle energy consumption E_{idle} when the machine is in standby state. The objective function of minimizing energy consumption is f_2, that is:

$$E_{\text{busy}} = \sum_{k=1}^{m} \sum_{i=1}^{n} \sum_{j=1}^{J_i} P_{ijk} X_{ijk} WP_k \tag{2}$$

$$E_{\text{idle}} = \sum_{k=1}^{m} \left(MF_k - \sum_{i=1}^{n} \sum_{j=1}^{J_i} P_{ijk} X_{ijk} \right) IP_k \tag{3}$$

$$f_2 = \min E_{\text{all}} = \min \left(E_{\text{busy}} + E_{\text{idle}} \right) \tag{4}$$

(3) Machine deviation D_{all} reflects the deviation between the actual processing machines of each operation in the rescheduling scheme and those in the original scheduling scheme. The objective function of minimizing machine deviation is f_3:

$$Md_{ij} = \begin{cases} 0, MS'_{ij} = MS_{ij} \\ 1, MS'_{ij} \neq MS_{ij} \end{cases} \tag{5}$$

$$f_3 = \min D_{\text{all}} = \sum_{i}^{n} \sum_{j}^{J_i} Md_{\text{ij}} \tag{6}$$

The constraints are established as follows:

$$F_{ijk} X_{ijk} = S_{ijk} X_{ijk} + P_{ijk} \tag{7}$$

$$F_{ijk} \leq S_{i(j+1)k} \tag{8}$$

$$\sum_{k=1}^{m} X_{ijk} = 1 \tag{9}$$

$$F_{ijk} \leq C_{\max} \tag{10}$$

$$MS_l^k = \max\left\{MF_{l-1}^k, F_{i(j-1)k'}, X_{ij}^k (T_{d1} + T_r)\right\} \tag{11}$$

$$\left(F_{ijk} - F_{i'j'k} - P_{ijk}\right) X_{ijk} X_{i'j'k} \geq 0, \quad S_{ijk} > S_{i'j'k} \tag{12}$$

$$\left(F_{i'j'k} - F_{ijk} - P_{i'j'k}\right) X_{ijk} X_{i'j'k} \geq 0, \quad S_{i'j'k} > S_{ijk} \tag{13}$$

$$S_{ijk} \geq 0, \quad F_{ijk} \geq 0, \quad P_{ijk} \geq 0 \tag{14}$$

Eq. (7) indicates that once an operation starts, it cannot be switched midway. Eq. (8) strictly constrains the process sequence relationship among operations of the same workpiece. In terms of resource exclusivity constraints, Eqs. (9), (12), and (13) jointly constitute machine capacity constraints, ensuring that each machine can only process one task at any time and each operation is assigned to only one processing unit. Eq. (10) indicates that the completion time of any operation must not exceed the makespan. Eq. (11) introduces the decision variable X_{ij}^k to indicate whether a specific operation is affected by machine failure disturbance. Eq. (14) is physical constraints to ensure that all variables related to start time, finish time, and duration satisfy non-negativity requirements.

3 Scheduling Solution Method Based on Improved Multi-objective Artificial Bee Colony Algorithm

3.1 Encoding and Decoding

The reasonable and refined design of the encoding mechanism serves as an essential prerequisite for realizing accurate and effective mapping from the complex scheduling solution space to the limited algorithm search space in intelligent optimization solving.

This paper constructs a two-layer hybrid encoding structure including Operation Sequence (OS) and Machine Sequence (MS). In this structure, the lengths of the OS segment and MS segment are both equal to the total number of operations of the workpieces to be processed, which are used to represent the processing order of operations and machine assignment decisions, respectively.

Figure 1 takes the scheduling of 4 types of workpieces on 5 machines as an example to illustrate the feasible solution form of this encoding scheme. Through specific decoding operations, the chromosome sequence can be transformed into a uniquely determined scheduling scheme, thereby realizing the digital representation of the complex scheduling process.

3.2 Population Initialization

The population initialization process should first respond to the real-time state constraints of the workshop. When a machine failure disturbance event occurs, the algorithm locks the interrupted operation of the workpiece currently being processed and re-inserts it into the available time window after machine repair; when an urgent order insertion disturbance event occurs, the system automatically integrates the new orders into the current unfinished queue. Subsequently, the initial solution set is generated using a random distribution function. The parameter x_{id} of the d-th dimension of the i-th food source x_i in the population can be expressed as:

$$x_{id} = L_d + \text{rand}(0, 1) \times (U_d - L_d), i = 1, 2, \cdots, N, j = 1, 2, \cdots, D \tag{15}$$

where, L_d represents the lower bound of the search space, and U_d represents the upper bound of the search space.

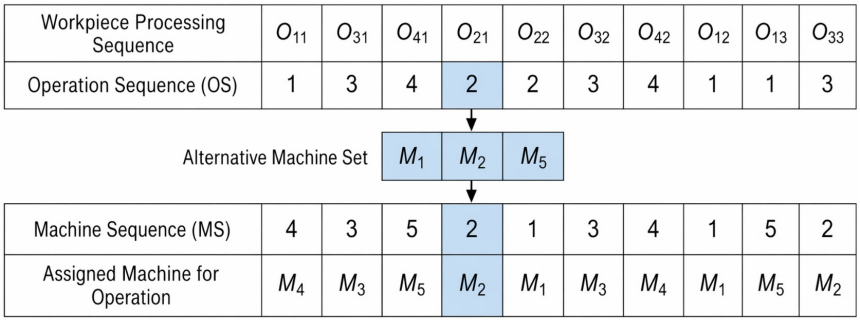


Figure 1. Schematic diagram of chromosome encoding

3.3 Employed Bee Operation

There is a one-to-one correspondence between employed bees and food source positions. For the OS part, this paper adopts the IPOX crossover strategy to adjust the processing order of workpieces, so as to avoid generating infeasible solutions. As shown in Figure 2, the specific steps are as follows: first, all workpieces to be processed are randomly divided into two complementary subsets, Subset 1 and Subset 2. When generating offspring C_1, the operations belonging to Subset 1 in parent P_1 are directly copied into C_1 according to their original positions in P_1; then, the operations belonging to Subset 2 in parent P_2 are identified and sequentially filled into the remaining vacant positions of C_1 according to their order in P_2. The generation process of offspring C_2 is similar.

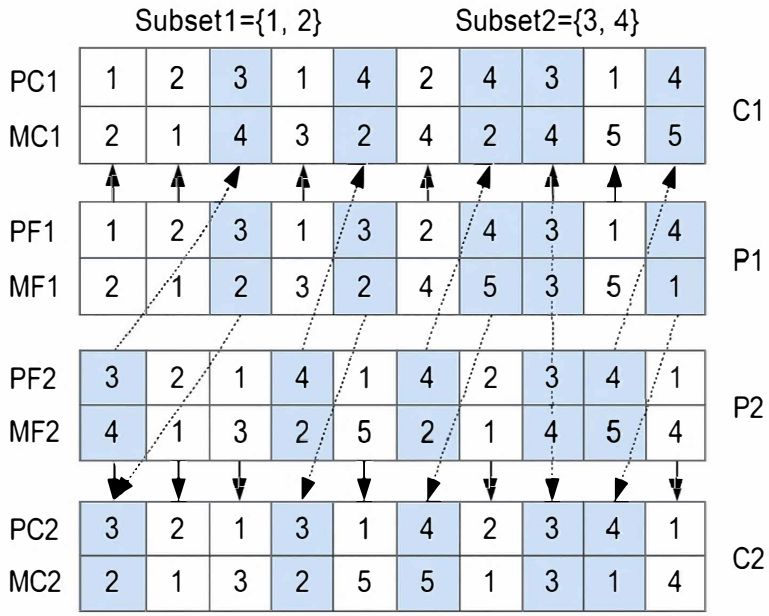


Figure 2. Schematic diagram of Improved Precedence Operation Crossover (IPOX) strategy

Note: PC = Parent Chromosome; PF = Progeny Offspring; MC = Machine Chromosome; MF = Machine Offspring.

For the optimization of the MS dimension, a uniform crossover strategy is introduced to reconstruct the machine assignment scheme.

As shown in Figure 3, this strategy first generates a binary mask vector with the same length as the chromosome as the crossover template. Under the control of this template, the algorithm performs bitwise selection for each operation of the workpiece: for positions where the mask value is "0", the offspring directly inherits the machine gene at the corresponding position of the parent; for positions where the mask value is "1", the gene exchange mechanism is triggered, that is, the machine encoding of parent MF_1 is passed to offspring MC_2, and conversely, the encoding of MF_2 is passed to MC_1. Through this position-based random exchange, the recombination of processing resources among offspring is realized.

In the employed bee search stage, in order to ensure the evolution quality of scheduling solutions, this paper adopts a two-level greedy selection mechanism combining Pareto dominance and crowding distance to update the population.

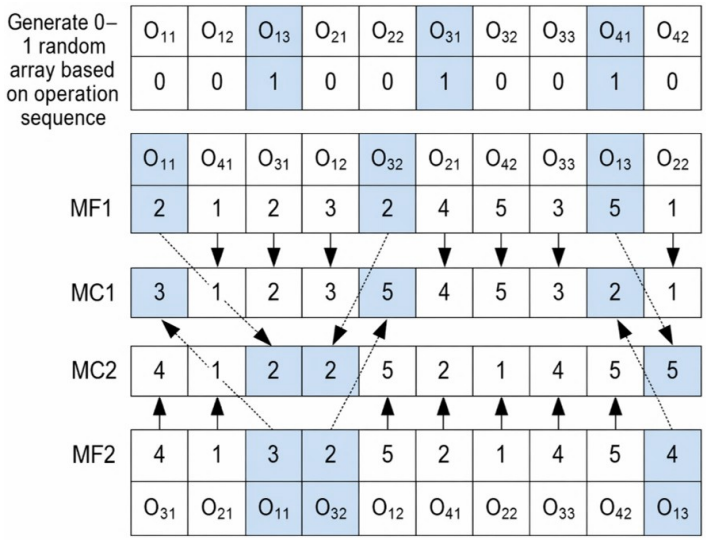


Figure 3. Schematic diagram of uniform crossover strategy

Note: MC = Machine Chromosome; MF = Machine Offspring.

The first level is dominance judgment: comparing the superiority between the newly generated solution x^* and the original solution x. If x^* dominates x, the new solution directly replaces the old solution, and the stagnation counter Trail_i is reset to zero; otherwise, if x dominates x^*, the original solution is retained and the counter is increased. The second level is diversity judgment: when the two are non-dominated, the crowding distance is introduced as the evaluation indicator.

The individual with a larger crowding distance is preferentially retained to maintain the distribution breadth of the solution set. In addition, to prevent the loss of elite solutions, all generated non-dominated solutions are stored in the external archive, and the specific state transition rule of the stagnation counter is shown in Eq. (16).

$$\mathrm{Trail}_i = \begin{cases} 0, x_i \text{ Updated} \\ \mathrm{Trail}_i + 1, x_i \text{ Not updated} \end{cases} \tag{16}$$

3.4 Onlooker Bee Operation

The main task of onlooker bees is to perform in-depth exploitation of high-quality solutions transmitted by employed bees. In order to balance the global exploration and local exploitation capabilities of the algorithm in a multi-objective environment, this section designs a probability selection mechanism based on Pareto rank and a variable step-size search strategy.

(1) Probability Selection Based on Pareto Rank

Since there is no single optimal solution in multi-objective optimization, the fitness cannot be directly calculated using function values. Therefore, this paper introduces Pareto dominance rank and crowding distance to construct the fitness evaluation system.

As shown in Eqs. (17) and (18), the higher the Pareto rank of an individual, the more solutions it dominates, the larger its fitness value fit_i, and the higher the probability P_i of being selected by the onlooker bee through the roulette wheel mechanism, ensuring that computational resources in the population are biased toward high-quality individuals.

$$P_i = fit_i \Big/ \sum_{i=1}^{N/2} fit_i \tag{17}$$

$$fit_i = \left(\mathrm{rank}(i) \exp\left(2m^i/N\right)\right)^{-1} \tag{18}$$

where, i represents the index of the selected individual, fit_i represents the comprehensive fitness value of the individual, $\mathrm{rank}(i)$ denotes the Pareto non-dominated level of individual i in the current population, and m^i is used to count the number of inferior solutions dominated by individual i.

(2) Variable Step-Size Neighborhood Search Mechanism

The traditional fixed step-size search method is difficult to balance convergence speed and the ability to escape from local optima. Therefore, this paper introduces a search threshold T to dynamically adjust the search step size.

When the number of searches for a solution is small, a small step-size strategy is adopted, in which only one pair of genes in the OS is exchanged to deeply explore the neighborhood of the current solution and accelerate convergence. When the solution has not been updated after multiple searches, a large step-size strategy is adopted, in which multiple pairs of operation genes are exchanged to expand the search radius, helping the algorithm escape from local optima and explore unknown solution space.

In this paper, insertion mutation is adopted for the OS layer, and random mutation within the alternative machine set is applied for the MS layer. Finally, the offspring individual with better fitness is retained and the external archive is updated.

3.5 Scout Bee Operation

The scout bee operation is mainly responsible for monitoring the population state and introducing new solutions when necessary. In order to ensure that excellent gene fragments are not lost during the evolution process, an elitism strategy is adopted to construct the next generation population. Specifically, the parent population of N food source positions generated by initialization is denoted as F. The parent population F and the offspring population C generated by crossover and mutation are merged and denoted as $F \cup C$. Then, fast non-dominated sorting and crowding distance calculation are performed on the mixed population. According to the principle of "Pareto rank priority and crowding distance secondary", the top N optimal individuals are selected to enter the next generation population P.

While updating the population, the algorithm monitors the activity of each food source in real time through the exploitation degree Trail_i. If the individual entering the next generation comes from the offspring population C, it indicates that evolution has occurred at this position, and the exploitation degree is reset to zero; if the individual still comes from the parent population F, it indicates stagnation, and the exploitation degree is accumulated. The update logic is shown in Eq. (19).

$$\mathrm{Trail}_i = \begin{cases} 0, x_i \in C, x_i \in F \\ \mathrm{Trail}_i + 1, x_i \in C, x_i \in F \end{cases} \tag{19}$$

When the exploitation degree Trail_i of a food source exceeds the maximum exploitation degree Limit, the solution is considered to have fallen into a local optimum trap.

At this time, the scout bee operation is triggered, the inferior solution is abandoned, and a new food source position is randomly generated in the solution space to replace it.

This mechanism introduces random disturbance and provides the possibility for the algorithm to escape from local optima in the later stage.

3.6 Processing Scheme Selection

When solving the dynamic scheduling problem, due to the mutual restriction among the three objective functions, the algorithm finally generates a set of non-dominated Pareto optimal solution sets. However, in the actual workshop production site, when facing sudden disturbances such as machine failures or urgent order insertions, the scheduling system must quickly and decisively select a specific execution scheme from the solution set.

At present, the linear weighted method is commonly used. The linear weighted method with n objectives can be defined as:

$$F(x) = \sum_{i=1}^{n} w_i f_{i'} \tag{20}$$

In Eq. (20), ω_i is the weight coefficient of the i-th objective function; f_i' is the normalized value of the i-th objective value f_i, which is calculated as follows:

$$f_i' = \frac{f_i^{\max} - f_i}{f_i^{\max} - f_i^{\min}} \tag{21}$$

where, $f_i^{\max}$ and $f_i^{\min}$ are the maximum and minimum values of f, respectively.

3.7 Procedure of Improved Multi-objective Artificial Bee Colony Algorithm

The IMOABC algorithm involves multiple new selection and search steps. The pseudocode of the algorithm is shown in Algorithm 1.

Algorithm 1 Algorithm pseudocode

Input: population size N, maximum iteration number Gen_{max}, maximum exploitation degree $Limit$, search threshold T
Output: Pareto front solution set Ω

$P, \Omega =$ PopInitialization() //Initialize population and construct external archive
while iteration $< Gen_{max}$ **do**
  for each employed bee X_i
    $V_i =$ HybridCrossover(X_i) //Perform Improved Precedence Operation Crossover (IPOX) and multi-point crossover
    ParetoGreedySelect(X_i, V_i) //Update solution and archive Ω based on dominance
  end for
  CalculateProb(P) //Calculate selection probability based on Pareto rank and crowding distance
  for each Onlooker bee (selected based on probability)
    if $Trail_i < T$ **then**
      $V_i =$ SmallStepSearch(X_i) //Perform small step-size mutation
    else
      $V_i =$ BigStepSearch(X_i) //Perform large step-size mutation
    end if
    ParetoGreedySelect(X_i, V_i) //Greedy selection to update current food source
  end for
  for each Scout bee X_i
    if $X_i > Limit$ **then**
      ReInitialize(X_i) //Randomly reset the food source trapped in local optimum
    end if
  end for
  ArchiveMaintain(Ω) //Remove individuals with smallest crowding distance to maintain archive size
  $Gen = Gen + 1$
end while
return Ω

4 Simulation Experiments and Result Analysis

4.1 Algorithm Verification and Performance Analysis

To fully verify the feasibility and effectiveness of the improved IMOABC algorithm proposed in this paper for practical workshop scheduling problems, comprehensive simulation experiments under both static scheduling and dynamic disturbance-based scheduling scenarios are reasonably established, and the experimental results are further summarized and analyzed in detail.

The relevant parameter settings of the IMOABC algorithm are as follows: population size N = 200, maximum iteration number $\mathrm{Gen}_{\max}$ = 200, crossover probability P_c = 0.8, mutation probability P_m = 0.2, maximum exploitation degree $\mathrm{Limit} = 20$, and strategy switching threshold $\mathrm{Threshold}$ = 5. The objective weights are set as follows: makespan weight w_1 = 0.64, energy consumption weight w_2 = 0.10, machine deviation weight w_3 = 0.26. The idle energy consumption of workshop machines is a random number in [0,1], and the processing energy consumption is a random number in [1,3].

The simulation test results based on benchmark instances are shown in Table 2, which further verify the advantages of the proposed algorithm in solving the MOFJSP problem. Among the 7 test cases, the scheduling schemes generated by this algorithm show obvious performance advantages in efficiency, energy consumption, and load balancing; among them, 6 cases achieve complete dominance over the solution sets of traditional algorithms.

Table 2. Results of benchmark instances

Instance Set	ID	$n \times m$	Traditional Artificial Bee Colony			Proposed Method		
			$C_{\max}$	E_{all}	D_{all}	$C_{\max}$	E_{all}	D_{all}
Kacem test set	K01	4×5	12.4	32.5	64.51	**11.3**	**32.0**	**63.12**
	K02	10×7	12.5	60.3	135.50	**12.4**	**60.2**	**135.44**
	K03	10×10	10.1	42.6	**87.95**	**8.4**	**41.8**	88.26
	K04	15×10	16.9	93.0	201.82	**13.7**	**91.7**	**198.89**
Brdata dataset	B05	6×10	51.8	153.5	322.24	**45.9**	**152.1**	**309.06**
	B06	10×6	36.3	142.8	305.82	**33.2**	**141.3**	**312.14**
	B07	15×8	**240.2**	**846.8**	2256.18	242.0	872.8	**2142.35**
	B08	15×4	192.4	676.9	1338.12	**188.5**	**675.9**	**1327.80**

4.2 Case Analysis of Dynamic Scheduling

In this paper, a complete Digital Twin simulation system oriented to the actual production of the reducer box flexible workshop is systematically constructed based on the mature AnyLogic simulation platform. Meanwhile, a reasonable two-dimensional comparative validation experiment is specially designed to conduct quantitative comparison and analysis. Under the unexpected machine failure disturbance scenario, the classic traditional "right-shift rescheduling strategy" is introduced as the baseline comparison method to intuitively analyze and discuss the practical optimization effect and application performance of the proposed IMOABC algorithm. Under the sudden urgent order insertion scenario, the common "original scheduling delay strategy" is selected as the control group, so as to comprehensively evaluate the machine load balancing capability and overall system robustness of the algorithm when coping with random sudden production loads.

The workshop is equipped with 8 machines and can produce 8 different types of workpieces. The relevant parameters are shown in Table 3. The processing energy consumption and idle energy consumption of machines are shown in Table 4.

Table 3. Processing time of operations in case study

Workpiece	Operation	Time of Operations							
		M_1	M_2	M_3	M_4	M_5	M_6	M_7	M_8
J_1	O_{11}			8			10	5	
	O_{12}				13				15
	O_{13}	9	20		12			22	
J_2	O_{21}			13		16		14	
	O_{22}		11		16				9
J_3	O_{31}	11		13		15			11
	O_{32}	6	12						19
	O_{33}	8	18				25		
	O_{34}				18	22		28	
J_4	O_{41}		14					17	
	O_{42}		13		15	19	18		
	O_{43}	20			19	22			21
	O_{44}	7		10		8	11	9	
J_5	O_{51}		22					17	23
	O_{52}			18				19	
	O_{53}	19		17	16	16			
J_6	O_{61}			16		17			15
	O_{62}	22					20	22	
	O_{63}		11				10		15
J_7	O_{71}			14	10	12	19		16
	O_{72}		8	13				11	
	O_{73}						14		16
	O_{74}		18					16	
	O_{75}	10	15	16			18	8	
J_8	O_{81}	7		6	5	10			
	O_{82}		18				17	18	
	O_{83}				10			12	18
	O_{84}		8	11	8			9	

Table 4. Processing and idle power of machines

Energy Type	M_1	M_2	M_3	M_4	M_5	M_6	M_7	M_8
Processing energy consumption WP_k	2.1	1.5	1.8	2.3	2.4	2.7	3.2	1.8
Idle energy consumption IP_k	0.6	0.4	0.3	0.4	0.5	0.3	0.6	0.5

To better illustrate the problem, the Gantt chart of the initial scheduling scheme is set as shown in Figure 4. On this basis, the initial scheduling scheme is rescheduled.

(1) Dynamic Scheduling Analysis under Machine Failure

Assume that machine M_4 fails at the 20th minute, and the duration is 10 minutes. The results of makespan $C_{\max}$, energy consumption E_{all}, and machine deviation D_{all} under the right-shift rescheduling strategy and the complete

rescheduling strategy based on the IMOABC algorithm are shown in Table 5.

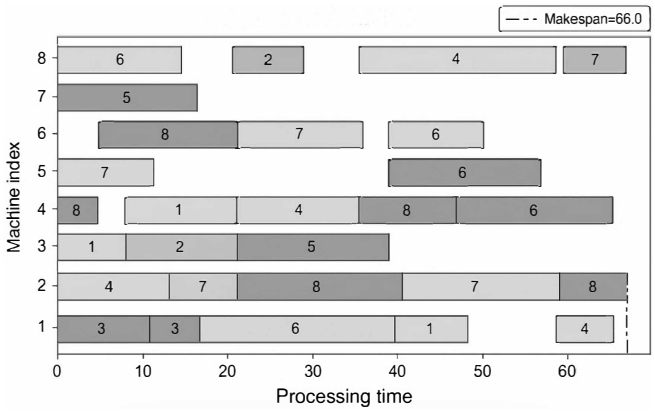


Figure 4. Gantt chart of initial scheduling scheme

Table 5. Performance metrics of scheduling schemes under machine failure

Scheduling Scheme	$C_{\max}$	E_{all}	D_{all}
Right-shift rescheduling	87	360.0	15.76
Complete rescheduling	68	307.0	9.11

The production scheduling Gantt charts generated based on the right-shift rescheduling strategy and the complete rescheduling strategy are shown in Figure 5 and Figure 6, respectively.

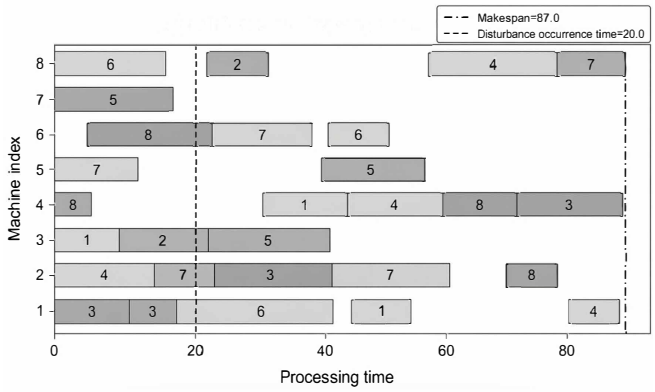


Figure 5. Gantt chart based on right-shift rescheduling strategy

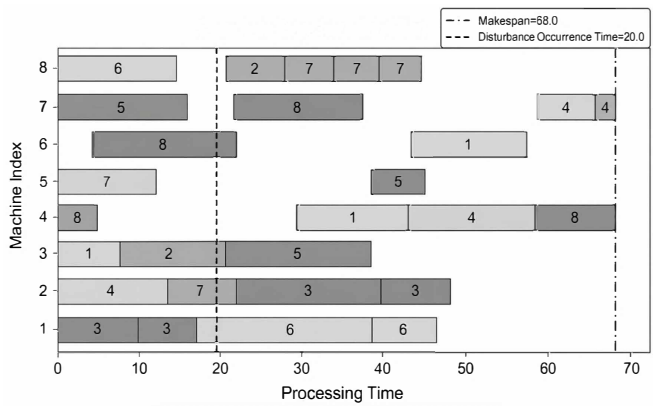


Figure 6. Gantt chart based on complete rescheduling strategy

The results show that, compared with the right-shift rescheduling strategy, the production scheduling scheme generated by the complete rescheduling strategy reduces the makespan $C_{\max}$ by 21.8% and decreases the energy consumption E_{all} by 14.7%. This result fully demonstrates that, in response to this machine failure disturbance event, the complete rescheduling strategy has better performance in improving scheduling robustness and maintaining the stability of the workshop system.

(2) Dynamic Scheduling Analysis under Urgent Order Insertion

An urgent order insertion occurs at the 20th minute, introducing two urgent workpieces J_4 and J_6 marked as NJ_4 and NJ_6. The workshop's original strategy schedules urgent tasks after ongoing workpieces, while the complete rescheduling strategy reoptimizes all unprocessed and urgent operations after the disturbance. Key indicators including makespan $C_{\max}$, energy consumption E_{all} and machine deviation D_{all} of the two scheduling methods are summarized in Table 6.

Table 6. Performance metrics of scheduling schemes under urgent order insertion

Scheduling Scheme	$C_{\max}$	E_{all}	D_{all}
Original rescheduling	149	471.0	21.40
Complete rescheduling	71	453.0	6.84

The detailed production scheduling Gantt charts corresponding to the two scheduling methods, which are separately generated based on the original scheduling strategy and the complete rescheduling strategy in this experiment, are visually presented in Figure 7 and Figure 8, respectively.

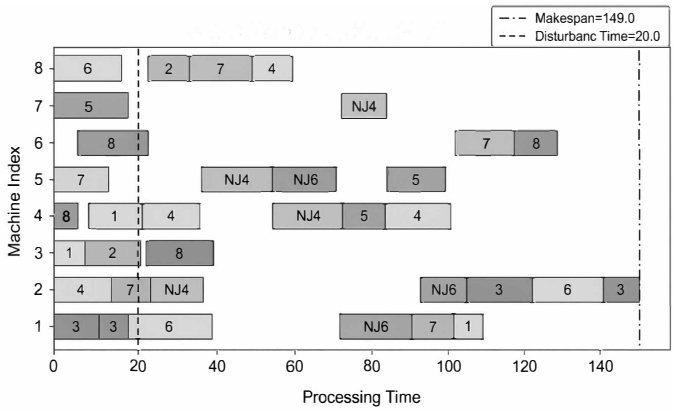


Figure 7. Gantt chart based on original scheduling strategy

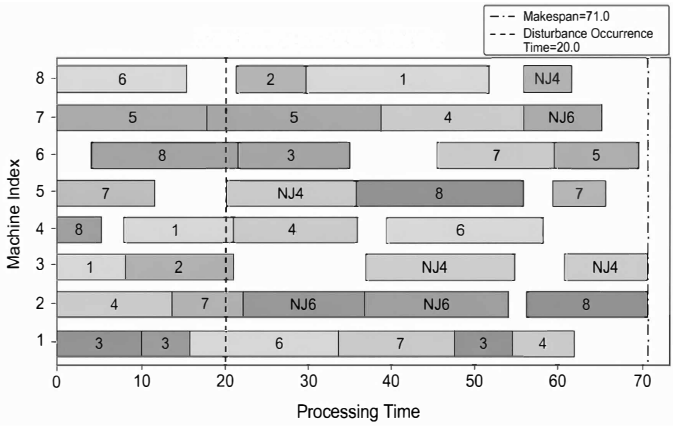


Figure 8. Gantt chart based on complete rescheduling strategy

Experimental results show that, compared with the conventional original scheduling strategy, the optimal production scheduling scheme generated by the complete rescheduling strategy effectively reduces the makespan $C_{\max}$ by 52.4% and decreases the overall energy consumption E_{all} by 3.8%. This distinct comparison result fully demonstrates that the optimized scheduling scheme obtained by adopting the complete rescheduling strategy can

better adapt to complex production conditions and satisfy the high flexibility requirements of intelligent modern workshop production systems.

5 Construction of Digital Twin Dynamic Scheduling Workshop Model

5.1 Framework Design of Digital Twin Workshop

The Digital Twin workshop relies on advanced information technology and intelligent manufacturing means to realize bidirectional mapping and real-time interaction between the Physical Workshop and the Virtual Workshop. Driven by workshop twin data, optimal integrated decisions are generated through analysis and prediction, thereby improving production management efficiency and system response capability. This paper designs a Digital Twin flexible job shop scheduling framework as shown in Figure 9, which mainly consists of four parts: simulation data layer, simulation function layer, simulation agent layer, and simulation scenario layer.

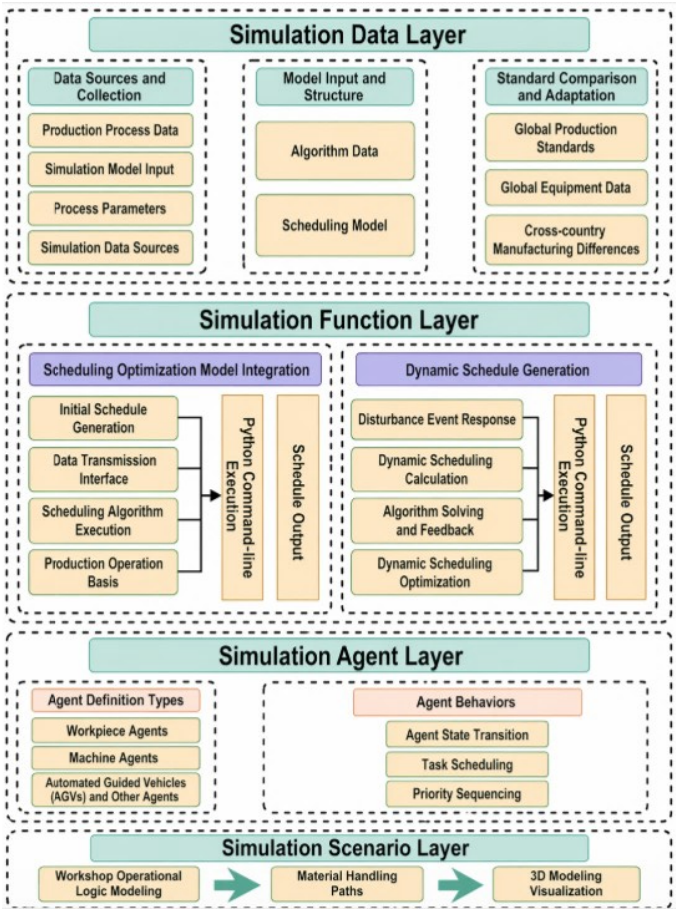


Figure 9. Digital twin flexible job shop scheduling framework

The scheduling framework includes the simulation data layer, simulation function layer, simulation agent layer, and simulation scenario layer: the data layer integrates inputs such as process data and models and uses Excel as the interface; the function layer, under dynamic disturbances, calls the IMOABC algorithm through the Python command line to complete data interaction and solution solving, and outputs executable scheduling results; the agent layer defines intelligent agents such as workpieces, machines, and Automated Guided Vehicles (AGVs), and configures attributes and behavior rules according to three types of objects: permanent entities, temporary entities, and system connections; the scenario layer completes the modeling of operation logic, logistics paths, and layout based on the process modeling library, material handling library, and space markup library of AnyLogic, and realizes visualization through 3D animation components.

5.2 Construction of Digital Twin Simulation Model Based on AnyLogic

AnyLogic is an object-oriented modeling and simulation platform based on standard Java, which supports multi-method integration, interactive simulation, and multi-experiment design, and provides data analysis tools and a

visual modeling interface. It has been widely used in manufacturing and supply chain scenarios. In this paper, a manufacturing workshop simulation model is constructed based on AnyLogic. By establishing agents, the attributes and behaviors of workshop entities are simulated to analyze the response characteristics of the workshop under dynamic disturbance events.

In this paper, a Digital Twin simulation model of the workshop is constructed based on AnyLogic, and the operation logic diagram of the workshop simulation model is designed as shown in Figure 10. P_i represents the i-th workpiece, and M_{kl} represents the l-th workpiece processed by the k-th machine. The working state of the processing machine is busy, and the idle state is idle.

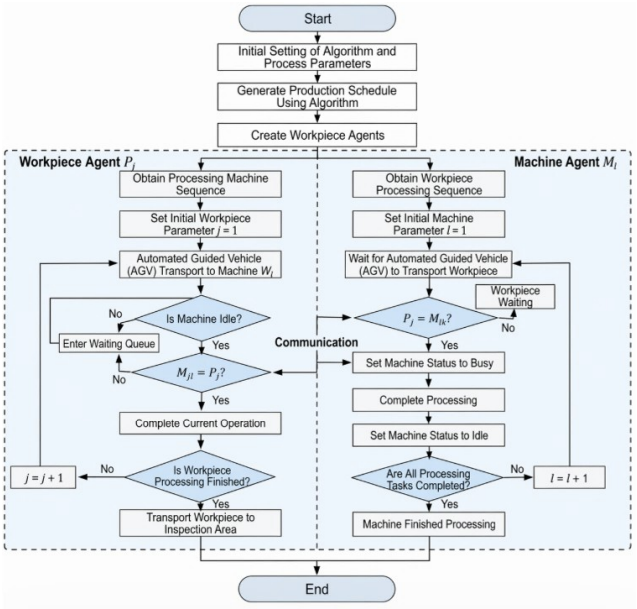


Figure 10. Operation logic diagram of workshop simulation model

First, the relevant parameters of processable workpieces in the workshop and the configuration of the scheduling algorithm are initialized. After real-time orders arrive, the system generates corresponding workpieces to be processed and obtains the original scheduling scheme.

For the workpiece agent P_i, the processing machines and processing times of each operation are obtained, and the workpiece is transported to the designated processing machine W_{ij} by AGV. At this time, the processing machine should belong to a certain machine M_k in the processing machine agent group. If machine M_k is currently idle and the next task of this operation is workpiece P_i, processing starts; otherwise, it enters the buffer for waiting. After the current operation is completed, if all processing tasks of the workpiece are finished, it is sent to the completion inspection area; otherwise, it is transferred to the processing machine of the next operation.

For the machine agent M_k in the processing machine agent group, the sequence of workpiece agents to be processed is first obtained. If workpiece P_i is the next task of machine M_k, processing starts; otherwise, the machine enters a waiting state. After completing all processing tasks of workpiece agents, the machine stops running; if not all tasks are completed, it continues to wait and process the next workpiece.

The interaction modules of the Digital Twin workshop model are constructed, and the parameter modules are shown in Figure 11, while the logical relationships among modules are shown in Figure 12. The interaction elements are mainly used for workshop environment initialization and real-time order storage; the production element module stores workpiece process data and basic parameters of the scheduling algorithm; the function element module is used for input, interaction, and type conversion of full-factor workshop data; the scheduling element module stores data related to workshop scheduling schemes; the collection element module stores workshop environment data and workpiece process data in a structured manner; the scheduling change element module is used for dynamic scheduling algorithm functions and rescheduling scheme storage; the AGV and conveyor elements store parameters related to AGVs and conveyors in the workshop. The main module types and their meanings are shown in Table 7.

The constructed workshop AnyLogic simulation model mainly includes four major areas: the incoming material area, the production processing area, the completion inspection area, and the finished goods storage area. The overall simulation effect of the workshop is shown in Figure 13 and Figure 14. It realizes the holographic mapping of the overall layout of the physical workshop and presents the execution process of workshop scheduling in a visual form, which is convenient for managers to make decisions and take direct actions.

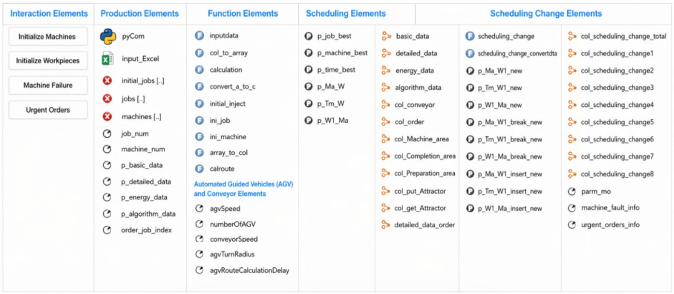


Figure 11. Parameters of digital twin model

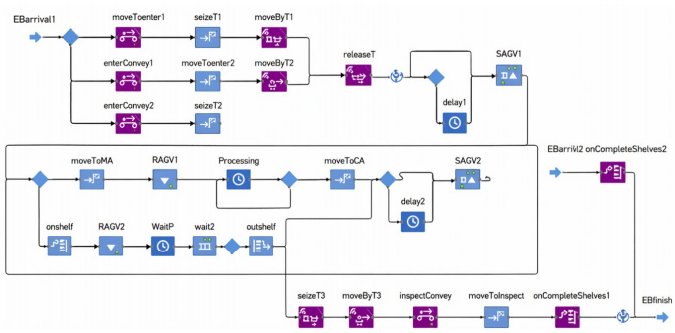


Figure 12. Logical relationships among modules in digital twin model

Table 7. Main module settings of digital twin model

Module Name	Module Type	Description
EBArrival1-2	Enter	Workpiece arrival
moveToEnter1-2	MoveTo	Workpiece enters production area
moveToMA	MoveTo	Transport workpiece to processing area
Processing	Delay	Workpiece processing
moveToCA	MoveTo	Workpiece moves to temporary storage after operation
onshelf	Rack Store	Workpiece enters buffer rack
outshelf	Rack Pick	Workpiece leaves buffer rack
WaitP	Delay	Workpiece waits for processing on rack
inspectConvey	Convey	Workpiece enters inspection process
moveToInspectA	MoveTo	Workpiece completes inspection process
onCompleteShelves1-2	Store	Workpiece enters finished buffer rack
EBfinish	Exit	Workpiece leaves after completion

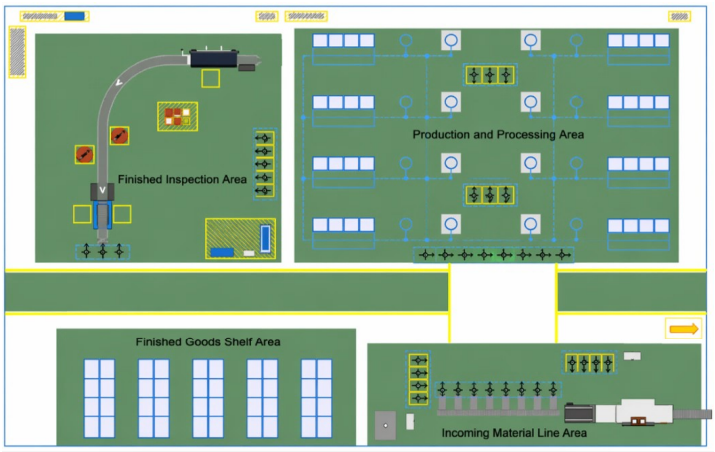


Figure 13. 2D view of digital twin simulation model operation

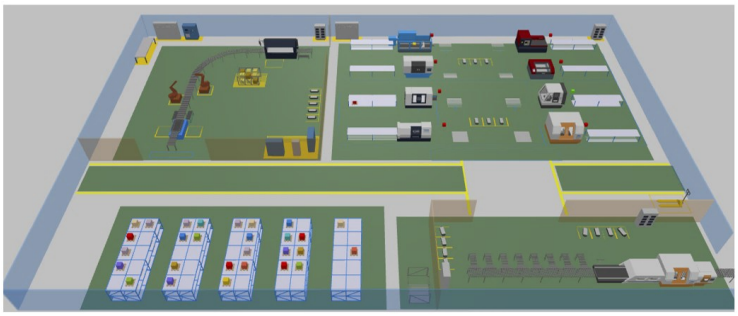


Figure 14. 3D view of digital twin simulation model operation

5.3 Analysis of Scheduling Optimization Effect of Simulation Model

5.3.1 Simulation analysis based on machine failure dynamic disturbance event

Assume that when the simulation model runs to the 50th minute, machine M_2 fails, and the repair time is 20 minutes. The complete rescheduling schemes generated by the right-shift rescheduling strategy and the IMOABC algorithm are obtained, and the rationality of these two scheduling schemes is verified through simulation. The processing time data under the two rescheduling strategies are collected and organized, and the Probability Density Function (PDF) and Cumulative Distribution Function (CDF) of the processing time intervals are calculated. The distribution of workpiece processing time is presented, and the specific data are shown in Table 8. The histograms of workpiece processing time under the right-shift rescheduling strategy and the complete rescheduling strategy are plotted, including the mean line and cumulative distribution curve, as shown in Figure 15.

Table 8. Workpiece processing time distribution under machine failure disturbance

Based on Right-Shift Rescheduling Strategy			**Based on Digital Twin Scheduling Strategy**		
Distribution Interval (min)	**Probability Density Function (PDF)**	**Cumulative Distribution Function (CDF)**	**Distribution Interval (min)**	**PDF**	**CDF**
[83.3,96.5)	2	0.2	[83.3,100.9)	2	0.2
[96.5,109.7)	1	0.3	[100.9,118.5)	1	0.3
[109.7,122.9)	1	0.4	[118.5,136.1)	2	0.5
[122.9,136.1)	1	0.5	[136.1,153.7)	1	0.6
[136.1,149.3)	1	0.6	[153.7,171.3)	2	0.8
[149.3,175.7)	1	0.7	[171.3,188.9)	1	0.9
[175.7,188.9)	1	0.8	[188.9,206.5]	1	1.00
[188.9,202.1)	1	0.9	-	-	-
[202.1,215.3]	1	1.00	-	-	-

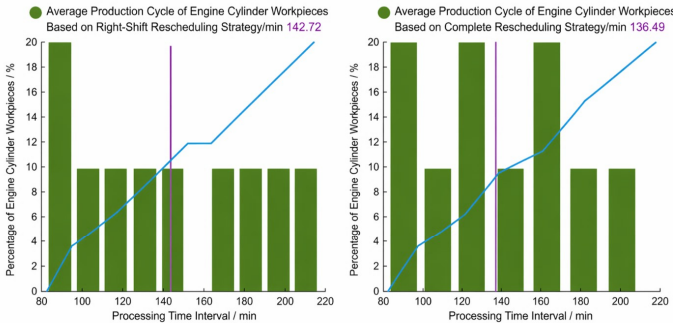


Figure 15. Workpiece processing time histogram under machine failure disturbance

From the comparison in Table 8, the workpiece processing time distribution under the digital twin scheduling strategy is more concentrated, indicating shorter overall processing duration, and thus higher solution quality. Combined with Figure 15, the average workpiece processing times under the right-shift rescheduling strategy and

digital twin scheduling strategy are 142.72 min and 136.49 min, respectively; compared with the traditional strategy, the digital twin scheduling strategy reduces the average processing time by 4.37%, thereby improving workshop production efficiency and ensuring the rationality and effectiveness of scheduling decisions, which verifies the solution performance of the proposed method for flexible job shop scheduling problems.

5.3.2 Simulation analysis under emergency order insertion dynamic disturbance

At the 30th minute of the AnyLogic simulation model, emergency workpieces J_1, J_2, and one of each J_8 are inserted. A complete rescheduling scheme based on the IMOABC algorithm is generated and simulated together with the scheduling scheme generated by the original scheduling strategy. The processing time distribution of workpieces is shown in Table 9. Workpiece processing time histograms under the original scheduling strategy and complete rescheduling strategy, including mean lines and CDF, are shown in Figure 16.

Table 9. Workpiece processing time distribution under emergency order insertion disturbance

Based on Original Scheduling Strategy			Based on Digital Twin Scheduling Strategy		
Distribution Interval (min)	**Probability Density Function (PDF)**	**Cumulative Distribution Function (CDF)**	**Distribution Interval (min)**	**PDF**	**CDF**
[74,92)	2	0.15	[72.4,85.6)	2	0.15
[92,128)	2	0.31	[85.6,112)	1	0.23
[128,146)	1	0.38	[112,125.2)	1	0.31
[146,164)	1	0.46	[125.2,138.4)	1	0.38
[164,182)	3	0.69	[138.4,151.6)	1	0.46
[182,200)	1	0.77	[151.6,164.8)	2	0.62
[200,218)	2	0.92	[164.8,178)	3	0.85
[218,236]	1	1.00	[178,191.2)	1	0.92
-	-	-	[191.2,204.4]	1	1.00

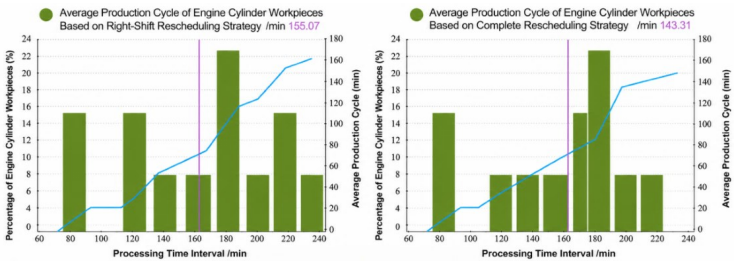


Figure 16. Workpiece processing time histogram under emergency order insertion disturbance

From Table 9, the workpiece processing time distribution under the digital twin scheduling strategy is more concentrated, indicating shorter overall processing time and higher solution quality. Combined with Figure 16, the average workpiece processing times under the original scheduling strategy and digital twin scheduling strategy are 155.07 min and 143.31 min, respectively; compared with the traditional strategy, the digital twin scheduling strategy reduces the average processing time by 7.58%, thereby improving workshop production efficiency and enhancing the rationality and effectiveness of scheduling decisions, further verifying the solution performance of the proposed method for flexible job shop scheduling problems.

6 Conclusion

Facing the insufficient scheduling response and information opacity of flexible job shops under dynamic disturbances, this paper proposes a scheduling optimization method based on digital twin simulation. A digital twin job shop scheduling framework is constructed, and a digital twin workshop model is established based on AnyLogic to achieve virtual-real linkage and workshop operation simulation. On this basis, complete rescheduling schemes are generated by the IMOABC algorithm. Comparing different strategies in terms of average workpiece processing time, order completion time, and processing time distribution shows that the proposed method can effectively shorten processing cycles, improve production efficiency, and enhance the disturbance resistance and production agility of workshop systems under disturbance conditions, providing support for stable and efficient workshop operation.

Author Contributions

Conceptualization, L.L.; methodology, L.L.; software, H.J.W.; validation, H.J.W. and Y.M.H.; formal analysis, H.J.W.; investigation, Y.M.H.; resources, L.L.; data curation, H.J.W.; writing—original draft preparation, Y.M.H.; writing—review and editing, L.L.; visualization, H.J.W.; supervision, L.L.; project administration, L.L. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

[1] W. He and D. H. Sun, "Scheduling flexible job shop problem subject to machine breakdown with route changing and right-shift strategies," *Int. J. Adv. Manuf. Technol.*, vol. 66, pp. 501–514, 2012. https://doi.org/10.1007/s00170-012-4344-4

[2] X. He, S. Dong, and N. Zhao, "Research on rush order insertion rescheduling problem under hybrid flow shop based on NSGA-III," *Int. J. Prod. Res.*, vol. 58, pp. 1161–1177, 2019. https://doi.org/10.1080/00207543.2019.1613581

[3] T. Küster, P. Rayling, R. Wiersig, and F. Pardo, "Multi-objective optimization of energy-efficient production schedules using genetic algorithms," *Optim. Eng.*, vol. 24, p. 447–468, 2021. https://doi.org/10.1007/s11081-021-09691-3

[4] R. Liu, R. Piplani, and C. Toro, "Deep reinforcement learning for dynamic scheduling of a flexible job shop," *Int. J. Prod. Res.*, vol. 60, no. 13, pp. 4049–4069, 2022. https://doi.org/10.1080/00207543.2022.2058432

[5] S. Luo, "Dynamic scheduling for flexible job shop with new job insertions by deep reinforcement learning," *Appl. Soft Comput.*, vol. 91, p. 106208, 2020. https://doi.org/10.1016/j.asoc.2020.106208

[6] S. Mahmud, R. Chakrabortty, A. Abbasi, and M. Ryan, "Switching strategy-based hybrid evolutionary algorithms for job shop scheduling problems," *J. Intell. Manuf.*, vol. 33, pp. 1939–1966, 2022. https://doi.org/10.1007/s10845-022-01940-1

[7] L. Wang, G. Zhou, Y. Xu, S. Wang, and M. Liu, "An effective artificial bee colony algorithm for the flexible job-shop scheduling problem," *Int. J. Adv. Manuf. Technol.*, vol. 60, pp. 303–315, 2011. https://doi.org/10.1007/s00170-011-3610-1

[8] L. Liu, K. Guo, Z. Gao, J. Li, and J. Sun, "Digital twin-driven adaptive scheduling for flexible job shops," *Sustainability*, vol. 14, no. 9, p. 5340, 2022. https://doi.org/10.3390/su14095340

[9] M. Sun, Z. Cai, and N. Zhao, "Design of intelligent manufacturing system based on digital twin for smart shop floors," *Int. J. Comput. Integr. Manuf.*, vol. 36, pp. 542–566, 2022. https://doi.org/10.1080/0951192x.2022.2128212

[10] Z. Liu, W. Chen, C. Zhang, C. Yang, and Q. Cheng, "Intelligent scheduling of a feature-process-machine tool supernetwork based on digital twin workshop," *J. Manuf. Syst.*, vol. 58, pp. 157–167, 2021. https://doi.org/10.1016/j.jmsy.2020.07.016

[11] Y. Wang and Z. Wu, "Model construction of planning and scheduling system based on digital twin," *Int. J. Adv. Manuf. Technol.*, vol. 109, pp. 2189–2203, 2020. https://doi.org/10.1007/s00170-020-05779-9

[12] E. Asorose and W. Adams, "Integrating lean six sigma and digital twins for predictive optimization in supply chain and operational excellence," *Int. J. Res. Publ. Rev.*, vol. 6, no. 2, pp. 1512–1527, 2025. https://doi.org/10.55248/gengpi.6.0225.0761

[13] Z. Chen, J. Zou, and W. Wang, "Digital twin-oriented collaborative optimization of fuzzy flexible job shop scheduling under multiple uncertainties," *Sādhanā*, vol. 48, p. 78, 2023. https://doi.org/10.1007/s12046-023-02133-z

[14] S. Luo, L. Zhang, and Y. Fan, "Dynamic multi-objective scheduling for flexible job shop by deep reinforcement learning," *Comput. Ind. Eng.*, vol. 159, p. 107489, 2021. https://doi.org/10.1016/j.cie.2021.107489

[15] R. H. Caldeira, A. Gnanavelbabu, and T. Vaidyanathan, "An effective backtracking search algorithm for multi-objective flexible job shop scheduling considering new job arrivals and energy consumption," *Comput. Ind. Eng.*, vol. 149, p. 106863, 2020. https://doi.org/10.1016/j.cie.2020.106863